MILITARY SPECIFICATION
PROPELLANT, COMPOSITE ADC-109

1. SCOPE

1.1 This specification covers one type of Polysulfide/Perchlorate composite propellant used for solid fuel rockets

2. APPLICABLE DOCUMENTS

2.1 The following documents, of exact issue shown, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

RR-S-366	21 Jun 66	Sieves, Standard; for Testing Purposes

Military

MIL-Q-9858	16 Dec 63	Quality Program Requirements
MIL-P-23702	10 May 63	Polysulfide, Polymer, Liquid for Rocket Motor, 3.0 Inch, Mark 32 Mod 0
MIL-Q-23703	24 Feb 65	Quinonedioxime, for Rocket Motor, 3.0 Inch, Mark 32 Mod 0
MIL-S-23704	26 Oct 64	Sulfur, Flowers of, for Rocket Motor, 3.0 Inch, Mark 32 Mod 0
MIL-D-23705	24 Feb 65	Diphenylguanidine, Symmetrical (DPG), Rocket Motor, 3.0 Inch, Mark 32 Mod 0

PUBLICATIONS

Bureau of Naval Weapons, Code Ident 10001

FSC 1340

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OS 9410	Aluminum Powder
OS 11327	Ammonium Perchlorate
OS 11645	Magnesium Oxide

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Composition. The propellant shall be composed of the materials shown in Table I, each of which meets the requirements of the applicable publication listed herein. The nominal ratio of the components is also shown in Table I. However, the supplier may vary the ratios, as necessary, to achieve the performance requirements of this specification.

Table I. Propellant Composition

Material	Purpose	Publication	Parts by Weight(nominal)
Polysulfide Polymer, Liquid	Fuel and Binder	MIL-P-23702	16.40
Quinonedioxime	Curing Agent	MIL-Q-23703	1.20
Sulfur, Flowers of	Curing Catalyst	MIL-S-23704	0.10
Diphenylguanidine	Curing Accelerator	MIL-D-23705	0.10
Ammonium Perchlorate* (as received)	Oxidizer	OS 11327	46.20
Ammonium Perchlorate* (after grinding)	Oxidizer	OS 11327	30.80
Dibutyl Phalate	Plasticizer		2.80
Magnesium Oxide	Curing Catalyst	OS 11645	0.60
Aluminum	Resonance Suppresent	OS 9410	1.80

*See Table II.

3.1.2 Particle size distribution. The particle size of the ammonium perchlorate shall be as shown in Table II.

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Table II. Grain Size Distribution for Ammonium Perchlorate Blend

Characteristic	Requirements Min	Max
As received AP retained on:		
USS 18	0	0
USS 50	3%	11%
USS 100	50%	82%
USS 140	85%	98%
USS 325	98%	100%
Ground AP % below:		
10 Microns	12	32
20 Microns	40	60
30 Microns	58	78.

3.2 Typical properties of cured propellant.

3.2.1 Physical properties. Cured Propellant ADC-109 exhibits the following typical physical properties at temperatures of 77°F when tested within 90-days from curing:

Density	0.062 lb/in^3
Hardness	70-90 Shore A
Tensile Strength	200-250 psi maximum
Elongation	18-45% maximum

3.2.2 Ballistic properties.

3.2.2.1 Temperature coefficient of pressure. The average temperature coefficient of pressure in the motor at constant k value, k equals 0.17 percent/°F. This value is determined on test motor firings over the temperature range of -40°F to +140°F. (The k value in a rocket motor is the ratio of propellant burning surface to the area of the nozzle throat.)

3.2.2.2 Heat of explosion. When tested with an adiabatic type calorimeter, the heat of explosion of uncured propellant is nominally 1,200 calories per gram.

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3.3 Workmanship. The propellant shall be a product free from foreign materials. It shall be uniform in quality, and manufactured in accordance with standard manufacturing procedures of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own or, any other facility and services acceptable to the Government. Inspection records of examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of material produced at one plant with no change in formulation or process. The raw materials of each propellant lot shall be from the same raw material lots, except that any one raw material may be from two lots, provided the two lots are from a single manufacturer's production.

4.3 Samples for acceptance tests. A sufficient amount of uncured propellant shall be taken from each lot to prepare cast sheets for tensile testing (see 4.4.2).

4.4 Acceptance tests. The following procedures shall be used to determine that the requirements of this specification have been met.

4.4.1 Composition. The contractor shall ascertain, prior to manufacture of this propellant, that all constituent materials have met the requirements of their applicable specifications.

4.4.2 Particle size distribution. The particle size of Ammonium Perchlorate shall be in accordance with Table I. Sieves used in testing shall be in accordance with RR-S-366.

4.4.3 Tensile test on cured propellant.

a. From each lot of propellant, a sample of propellant shall be poured (by gravity) into a single or multiple cavity mold large enough to cut five or more acceptable tensile test samples (see Figure 1). The propellant shall be cured in the mold for the same time and under the same temperature conditions as the remainder of the batch of propellant.

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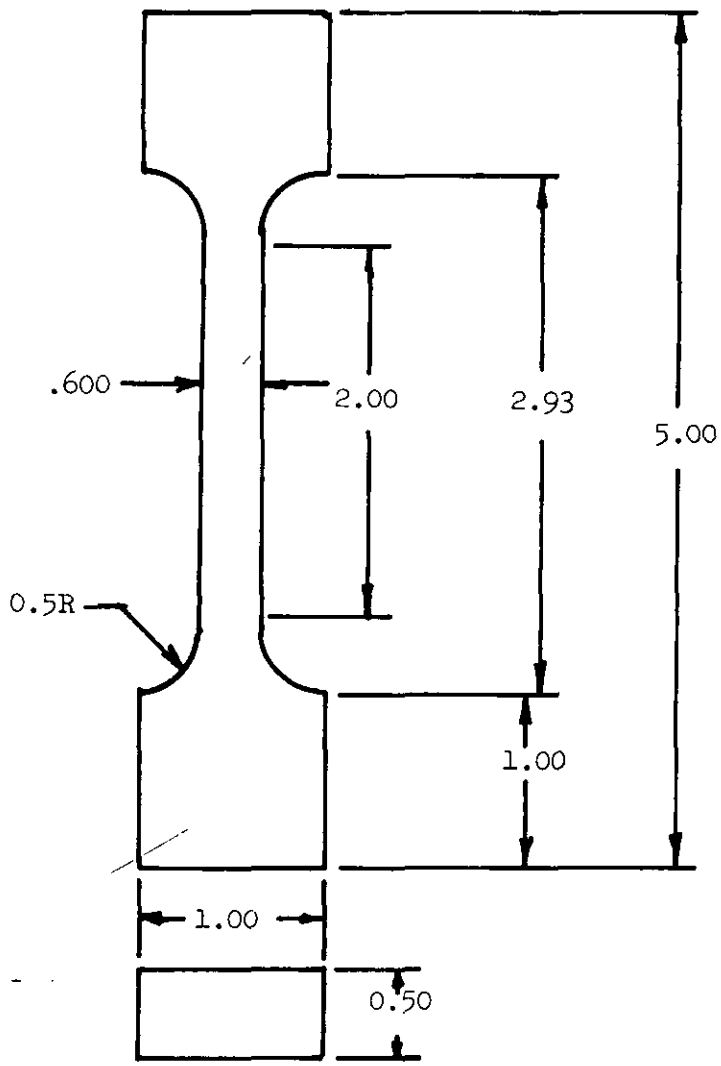


FIGURE 1. TENSILE TEST SPECIMEN

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b. At random, five test samples shall be cut (see Figure 1) from the cured test sheet above. Material from the top one-inch of the test mold shall not be used. Test samples having visual cracks or voids shall not be used.

c. The minimum cross-sectional area in the approximately 2-inch long test section of each specimen shall be determined by measuring all dimensions to the nearest 0.001 inch.

d. Each sample shall be temperature conditioned to a temperature of 75 to 79 degrees Fahrenheit (°F) for at least two-hours.

e. A temperature-conditioned sample shall be attached to the jaws of the tensile test machine. Within 15-minutes, the testing jaws shall be separated at a rate of 2-inches per minute.

f. The readings from the graph of the tensile test machine shall be recorded. The maximum tensile strength and the elongation of maximum load (strain at maximum stress) shall be calculated. The test results shall meet the requirements of 3.2.1.

4.4.4 Rejection criteria. If a test specimen fails to meet any of the acceptance tests required by this specification, the lot represented by the sample shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Not applicable.

6. NOTES

6.1 Intended use. This material is intended for use as propellant for rocket motors. This propellant is composed essentially of a dispersion of Ammonium Perchlorate Oxidizer in a matrix of Polysulfide synthetic rubber which serves the dual function of fuel and binder. The propellant is formed by casting a fluid mixture of the ingredients in a motor case and heating the assembly to cure the mixture to an elastic solid. Internal burning surfaces are formed by a mandrel or core which is removed after the propellant is cured.

6.2 Ordering data. Procurement documents should specify, but not be limited to the following:

a. Title, number and date of this specification.

b. Use of MIL-Q-9858.

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6.3 Definitions.

6.3.1 Batch. A batch is defined as that quantity of material which has been subjected to one or more chemical or physical processes (or combinations thereof) intended to produce a desired product having substantially uniform characteristics.

6.4 General safety precautions. The loading, assembly and handling of the item covered by this specification, and the subassemblies thereof, involve hazardous operations and, therefore, require suitable explosives safety precautions. Use of this specification will not be construed as to relieve the supplier or manufacturer of his responsibility for the safety of his operations. Listed below are certain minimum provisions which a supplier or manufacturer should observe in order to fulfill his responsibility for safety.

6.4.1 All loading operations should be conducted in a neat and orderly manner.

6.4.2 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosives operations, such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

6.4.3 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which affect functioning, should insofar as possible, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.4.4 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.4.5 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance with the American Table of Distances (ATD) or other applicable safety standards; and, while in process, in safety lockers and chests in in loading rooms, or in adequate ready or service magazines located in accordance with intraplant distances when outside of loading rooms. For Navy managed explosives loading plants, the provisions of the Armed Services Explosives Safety Board, covering quantity distance relations for explosives, will apply.

6.4.6 Proper care must be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In this connection, the precautionary measures in the following paragraphs should be observed.

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6.4.6.1 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.4.6.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.4.6.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of, or put into, the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.4.6.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and worker's conductive shoes. Worker's clothing of a type to minimize the accumulation of static charges should be employed. Fabrics such as silk and nylon, which promote static generation should be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are usually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low energy type electric primers, detonators and squibs. The latter types of items should have the free ends of lead in bare non-insulated Aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of these items from induced electric current generated by sources such as lightning, static, radiations from communications apparatus, radar, or high frequency heat apparatus, etc. Where necessary for safety, humidity of work rooms should be appropriately increased, as required to lessen electrostatic effects, but without inducing excessive moisture absorption by any of the components of the item being loaded.

6.4.6.5 Protect all explosive operations from effects of electric current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed, for example, from a defective electric soldering iron through a grounded contact. All electric type primers, detonators or squibs provided with wire leads should have the free ends bared and twisted together to short circuit

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each unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting clips, or other devices should be employed during manufacturing operations involving electric primers, detonators or squibs.

6.4.6.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impreganted garments to protect personnel against burns, poisoning and associated industrial hazards.

6.4.6.7 Allow no fires or exposed electrical or other sparcking equipment, and little or no flammable material to be present in loading, handling, and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.4.6.8 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping, and equipment or the items loaded; also clean-up any spilled material immediately.

Custodian:

Air Force - 13

Preparing Activity:

Air Force - 13

Review Activities:

Air Force - 70

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